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Skill : WEEK 11

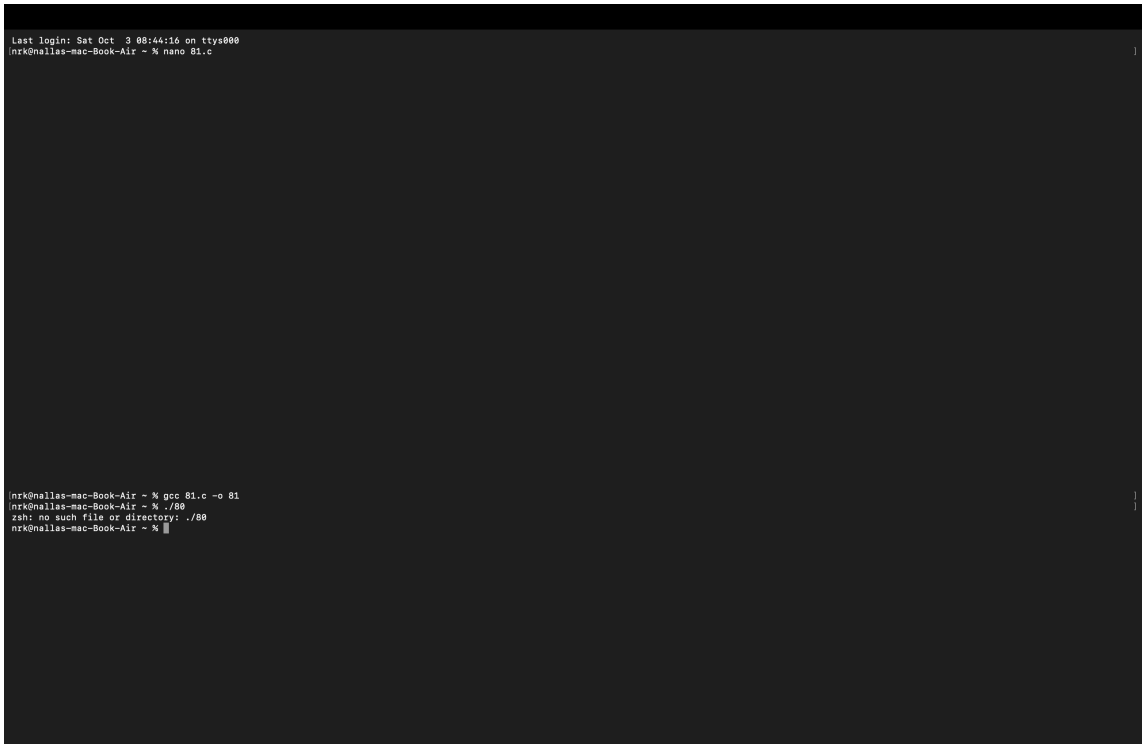
Task -1:

To Parse Input Redirection, Open Files, Redirectstdin, Handle MissingFiles, Restore

Streams,Test File Inputs

Code:

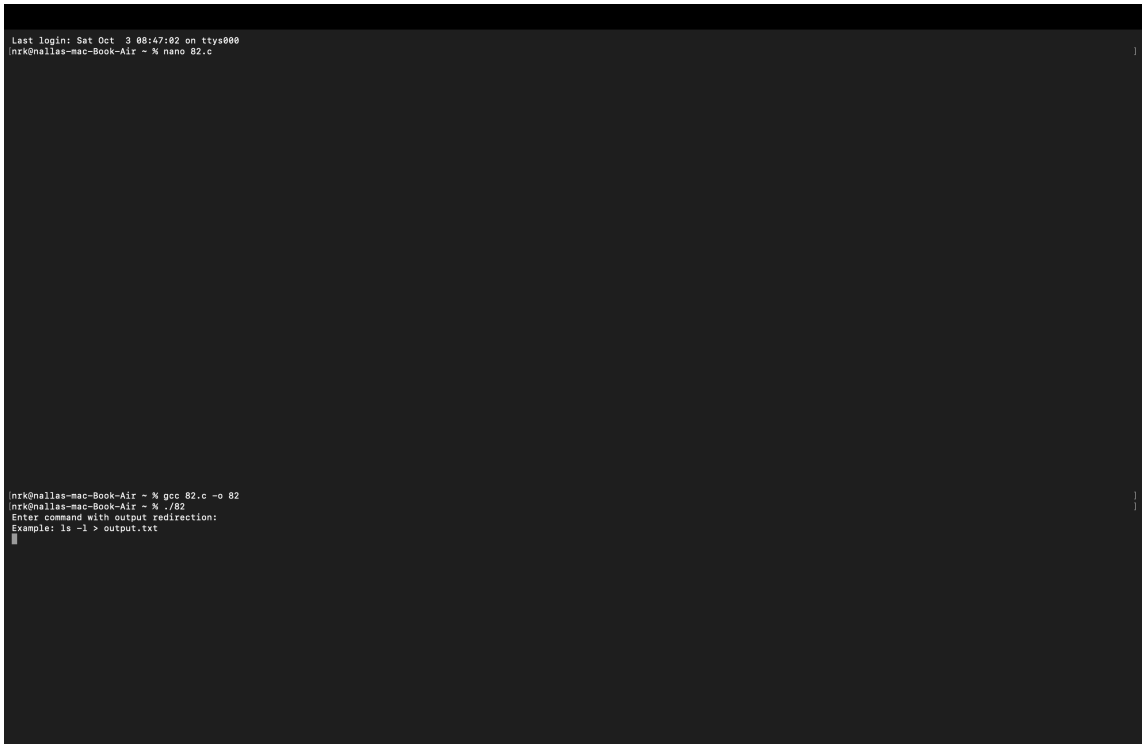




2.To Parse OutputRedirection, Create OutputFiles, Redirect stdout,Handle Permissions, Verify Output Data, Test Redirection

Code:

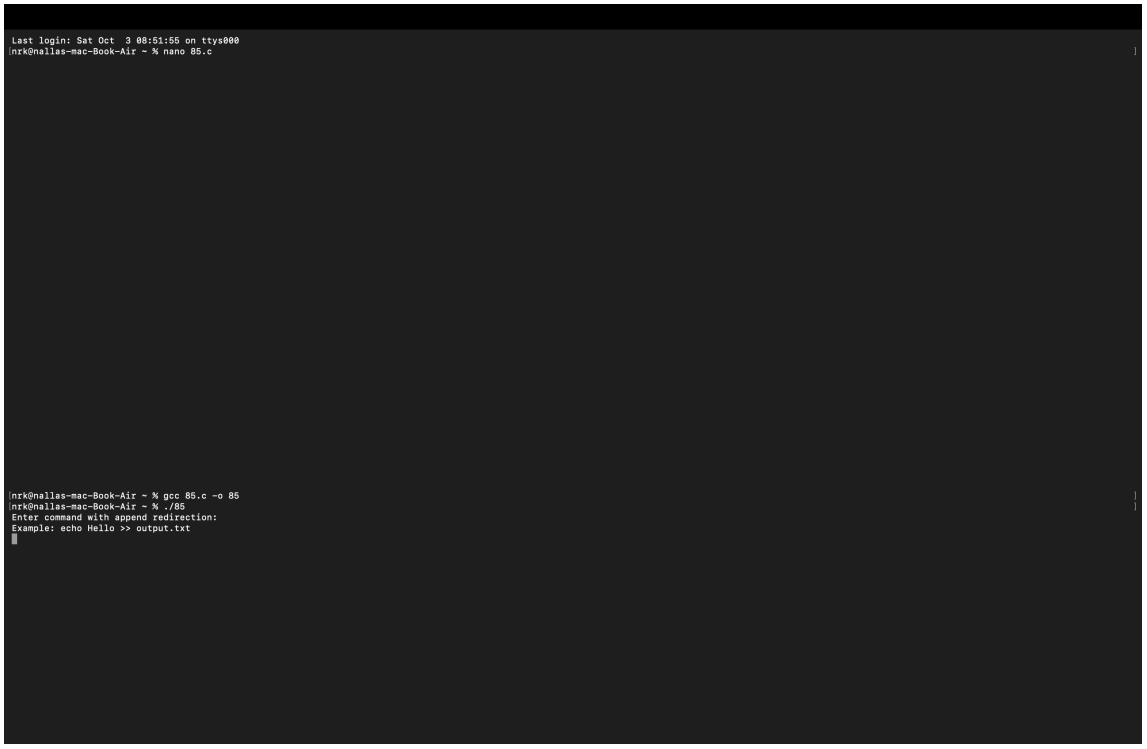




3) To Parse Append Operator,Open Files in Append Mode, Redirect Streams,Preserve Existing Data, Validate Appends, Test Scenarios



Output:



4)To Parsestderr Redirection, RedirectError Streams, CaptureError Output, VerifyFile Contents, Handle Failures, Test Error Cases

Code:

```
 UW PICO 5.09                          File: 86.c                          Modified

        while (token != NULL && count < 99) {
            args[count++] = token;
            token = strtok(NULL, " \t");
        }

        args[count] = NULL;

        if (count == 0) {
            fprintf(stderr, "Error: Empty command.\n");
            exit(1);
        }

        // Execute command
        execvp(args[0], args);

        // If execvp fails, error goes to redirected stderr
        perror("execvp failed");
        exit(1);
    }

    // Parent process
    int status;

    waitpid(pid, &status, 0);

    printf("\nCommand execution completed.\n");

    // Verify the error output file
    FILE *fp = fopen(file, "r");

    if (fp == NULL) {
        perror("Error reading error output file");
        return 1;
    }

    printf("\n--- Captured Error Output ---\n");

    char buffer[256];

    while (fgets(buffer, sizeof(buffer), fp) != NULL) {
        printf("%s", buffer);
    }

    fclose(fp);

    printf("\n--- Verification Complete ---\n");

    // Check command status
    if (WIFEXITED(status)) {
        printf("Exit Status: %d\n", WEXITSTATUS(status));
    } else {
        printf("Command terminated abnormally.\n");
    }

    return 0;
}

^G Get Help    ^O WriteOut    ^R Read File    ^Y Prev Pg    ^K Cut Text      ^C Cur Pos
^X Exit        ^J Justify     ^W Where is     ^V Next Pg    ^U UnCut Text    ^T To Spell
```

Output:

```
[nrk@nallas-mac-Book-Air ~ % nano 86.c                                                ]
[nrk@nallas-mac-Book-Air ~ % gcc 86.c -o 86                                           ]
[nrk@nallas-mac-Book-Air ~ % ./86                                                     ]
Enter command with stderr redirection:
Example: ls nonexistent 2> error.txt
1
Error: stderr redirection operator '2>' not found.
nrk@nallas-mac-Book-Air ~ %
```